

IIUoT at TREC 2024 Biomedical Generative Retrieval Track

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In this paper, we present our approach used in the retrieval and generation phases for the Biomedical Generative Retrieval (BioGen) task. BioGen is a specialized task designed to enhance the reliability of large language models (LLMs) in the biomedical domain by ensuring accurate reference attribution for generated statements. Our comprehensive pipeline for this task includes *DocID generation*, *relevant document retrieval*, and *response generation*. Specifically, we use a fine-tuned pre-trained T5 model for relevance-based DocID generation, where document IDs, abstracts, and queries are used as inputs to generate DocIDs that effectively capture document-query relevance. This step enhances document representation for large-scale retrieval tasks. For response generation, we first retrieve the top 10 most relevant document IDs for each query. We then use the GPT-4o API model to generate responses based on the retrieved documents. GPT-4o takes the query, the document IDs, and abstracts of the candidate documents to produce an answer that aligns with the query’s intent and requirements, providing a comprehensive and contextually relevant response for summarization tasks.

1 INTRODUCTION

The Biomedical Generative Retrieval (BioGen) task builds upon the existing challenges of retrieval and response generation in the biomedical domain, focusing on enhancing the reliability of large language models (LLMs) by ensuring accurate reference attribution. Participating groups are provided with 40 biomedical topics and a stable version of PubMed documents to generate responses with LLMs that include up to three attributions (cited references) per statement, limited to 50 documents per answer. Each document should be referenced as PMIDs in square brackets.

The BioGen task aims to improve LLM responses by incorporating reliable and well-supported references, with an emphasis on real-time adaptation to evolving contexts and user needs. Specifically, the task involves generating LLM output that provides accurate and well-supported answers for each biomedical query while attributing sources correctly.

In short, BioGen advances research in conversational biomedical retrieval by emphasizing personalized, context-aware responses with reliable attributions. To delve deeper into this task, we focus on the following key parts:

- **Part 1 : *Evaluating Answer Alignment with Questions and Evidence Support*** We evaluate whether incorporating document IDs and abstracts during DocID generation enhances relevance and retrieval effectiveness. This part aims to ensure that the generated answers align well with the questions and are backed by appropriate evidence.
- **Part 2: *Answer Quality and Completeness*** We assess whether LLMs are more effective in generating reliable biomedical responses compared to traditional retrieval-based methods. This part also considers the computational cost and the justification for employing LLMs for large-scale biomedical retrieval tasks, focusing on the quality and completeness of the answers generated.

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This paper outlines our approach and results for the BioGen task. Our approach can be summarized as a robust three-stage pipeline, encompassing the following key components: (1) DocID generation, (2) relevant document retrieval, and (3) response generation. In the following section, we will provide a brief overview of each stage of our approach.

2 METHODOLOGY

This paper introduces our method, a generic framework for document ID construction and prefix-oriented ranking optimization that can be applied to any encoder-decoder architecture, enhancing the performance of generative retrieval models. An overview of our method framework is provided in Figure 2. Initially, the generative model M is treated as a dense encoder and undergoes fine-tuning with a relevance-based objective. Once trained, our method utilizes Residual Quantization (RQ)[1] to generate a unique identifier for each document. Following this, a sequence-to-sequence pre-training approach is employed, which uses pseudo queries generated from the documents for model pre-training. Subsequently, a novel rank-oriented fine-tuning process is introduced to further refine the parameters of model M . The following two sections elaborate on the motivations and methodologies behind the two main components of our method: prefix-oriented ranking optimization and relevance-based document ID construction. A comprehensive description of the optimization pipeline can be found in Section 2.2.

2.1 Relevance-Based DocID Construction

Generative retrieval models often utilize a two-phase optimization strategy. In the first phase, document IDs are generated using methods such as hierarchical k-means [4][5] clustering or by deriving discriminative textual features from the documents. In the second phase, the model is trained using either cross-entropy loss or a learning-to-rank loss function, based on the fixed document IDs established earlier. Since these document IDs are static during this stage, they may become a performance bottleneck, impacting the efficiency of generative retrieval models.

We assert that the design of document IDs is critical for two reasons: firstly, they should reflect the inherent similarity between documents by assigning similar IDs to similar content. Secondly, given the hierarchical nature of beam search in generative retrieval, these IDs should also convey a hierarchical structure. Importantly, the notion of similarity here is nuanced and depends on specific queries, which differs from the general linguistic similarities often considered in natural language processing. To address these issues, we propose a relevance-focused method for document ID initialization, which aims to incorporate both the query-document relevance and the necessary hierarchical structure, thereby improving the effectiveness of generative retrieval tasks.

Generative retrieval models can also function as dense encoders. To capture relevance-based similarities among documents, we design an optimization process inspired by dense retrieval models, utilizing an encoder-decoder architecture within model M . Specifically, the document content is passed into the encoder, while a special start token is provided as input to the decoder. The document representation is then derived from the first contextualized output embedding of the decoder:

$$\mathbf{d} = \text{Decoder}(s_0; \text{Encoder}(\mathbf{d})) \in \mathbb{R}^D, \quad (1)$$

Where s_0 is the start token. By following a similar process for queries, we also determine their representations. To optimize model M , we employ the MarginMSE loss, along with multi-stage negative sampling[7] as detailed in Section 2.2.1.

Hierarchical k-means, commonly used for document ID construction, does not directly minimize the distortion error between the original and approximated representations. It has been observed that there is a significant inverse correlation between information retrieval metrics like MAP and the distortion error, especially for large-scale datasets. Based on this insight, we employ quantization-based techniques specifically designed to minimize this distortion error. Among various quantization algorithms, we choose Residual Quantization (RQ) due to its distinct benefits. Specifically, (1) its recursive process captures the hierarchical structure of documents, which aligns well with the beam search strategy in generative retrieval, and (2) compared to methods like product quantization (PQ), it requires shorter DocIDs to achieve high performance, resulting in memory and time efficiency during inference.

Using M as our dense encoder, we calculate the representation $\mathbf{d}$ for each document d . We then apply RQ to optimize the token embedding table $\{\mathbf{E}_i\}_{i=1}^L$ and determine the optimal DocID $\mathbf{c}_a = [c_i^d, \dots, c_i^d]$ for each document d . After optimization, each document representation $\mathbf{d}$ can be approximated using a sequence of token embeddings as:

$$\mathbf{d} \approx \sum_{i=1}^L \mathbf{E}_i [c_i^d] .$$

The trained model M and the embedding tables $\{\mathbf{E}_i\}_{i=1}^L$ will be used as the initial weights for further optimization stages in generative retrieval.

2.2 Optimization Details

Our optimization process consists of three phases: (1) DocID Initialization, (2) Seq2seq Pre-training, and (3) Rank-oriented Fine-tuning.

2.2.1 DocID Initialization. In this phase, the model M is considered as a dense encoder, and optimization is performed using a multi-stage training strategy. Initially, BM25 [3] is used to sample the top K documents (with $K = 100$) for each query, and the model is trained with the MarginMSE loss function. Once the initial training is completed, dense representations d are obtained from model M for each document. For each query q , nearest neighbor search is used to retrieve the top K documents, and the model is then retrained using the same loss function. Residual quantization (RQ) is then applied to obtain the DocID for each document. The trained model from this phase is referred to as M_0 , and the embedding tables $\{\mathbf{E}_i\}$ are used as initial weights for the next phase.

2.2.2 Seq2seq Pre-training. To provide the model M with a comprehensive understanding of the dataset, a seq2seq pre-training phase is introduced. Instead of using the document d as input for predicting semantic tokens, we use pseudo queries associated with each document for DocID prediction. Specifically, the doc2query[6] model generates multiple pseudo queries for each document. The model is then optimized using cross-entropy loss, with tokens from the relevant DocIDs used as ground-truth labels. The model trained in this phase is denoted as M_1 .

2.2.3 Rank-oriented Fine-tuning. In the final phase, the model is further optimized using a rank-oriented fine-tuning strategy. We employ a pairwise loss function to improve the model’s ranking performance. Negative sampling[7]s plays a crucial role in this phase, as it helps enhance the effectiveness of the ranking model. A multi-stage training approach is used to gradually improve the model’s capacity and refine the selection of negative samples across different stages, ultimately leading to a more effective ranking model.

3 EXPERIMENTS

3.1 Experiments Settings

We evaluate our retrieval models on the BioGen dataset, which uses the latest annual baseline snapshot of PubMed, covering content up through the end of 2023. This dataset consists of a pre-processed set of 20,727,695 PMIDs, originally in XML format, representing abstracts from the 2023 snapshot. To improve the model’s understanding and generalization performance on the medical dataset, we randomly selected 81,652 PMIDs during data preprocessing, extracting only the title and abstract sections from the XML data for training. Due to the absence of corresponding query data samples in the PubMed dataset, we utilize the title texts as the query dataset for model training. Since this dataset has not been previously used for training and lacks individual sample labels, we employ unsupervised semantic similarity metrics to assess the relevance of the retrieved samples to each query. Specifically, we report MRR@10, Success Rate@k as evaluation metrics. These metrics provide insights into how well the retrieved content aligns semantically with the queries, given the absence of labeled relevance data.

3.1.1 Implementation Details. We use the pre-trained T5-base model as the foundation for our generative retrieval model. For DocID initialization, we utilize residual quantization (RQ) from Faiss. The length of the document ID (L) is set to 32, and the vocabulary size (V) is 256. During sequence-to-sequence pre-training, the T5-large doc2query model generates 3 pseudo queries per document. The doc2query model is trained on the PubMed dataset using cross-entropy loss. Optimization is performed with the Adam optimizer, using linear scheduling and a warm-up ratio of 4.5% of the total learning steps. For the DocID initialization and rank-oriented fine-tuning stages, the learning rate is set to 0.0001, with 120 epochs and a batch size of 64. For sequence-to-sequence pre-training, the learning rate is set to 0.001, with 15,000 steps and a batch size of 256. All experiments were conducted using two 40GB A100 GPUs.

3.1.2 Baselines. As no previous research has applied generative information retrieval models to the PubMed dataset for training and evaluation, our experiments represent the first attempt in this domain. Consequently, this study does not include baseline model comparisons in the conventional sense. However, to verify the effectiveness and performance of our model, we compare it with BM25 using the following metrics:

- **Mean Reciprocal Rank (MRR@10):** Evaluates the ranking quality of the retrieved documents.
- **Success Rate @ 5:** Measures the proportion of times the correct document appears within the top 5 results.
- **Success Rate @ 10:** Measures the proportion of times the correct document appears within the top 10 results.

3.2 Experiment Results

Given the lack of labeled query-document pairs in the PubMed dataset, we utilize the aforementioned metrics to assess the effectiveness of our model’s retrieval performance. Specifically, we compare the performance of our model with BM25 using MRR@10, Success Rate @ 5, and Success Rate @ 10 to evaluate how well the generated passages align semantically with the actual content of the cited literature.

3.2.1 Main Results. The experimental results on the BioGen dataset demonstrate that our generative information retrieval model effectively returns content from relevant citations in response to medical queries, achieving significantly higher scores compared to BM25 on all evaluation metrics. For instance, with 1000 queries, our model attained an MRR@10 of 0.573, a Success Rate @ 5 of 0.664, and a Success Rate @ 10 of 0.695, whereas BM25 only achieved 0.362, 0.433, and 0.498, respectively. These results indicate that, despite the lack of traditional baseline comparisons, our approach exhibits substantial advantages in terms of semantic consistency and relevance.

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Table 1. Performance Comparison between Our Model and BM25

Query Count	Model	MRR@10	Success Rate @ 5	Success Rate @ 10
100	Our Model	0.478	0.590	0.650
	BM25	0.333	0.380	0.460
200	Our Model	0.551	0.660	0.715
	BM25	0.359	0.425	0.510
500	Our Model	0.572	0.664	0.700
	BM25	0.355	0.430	0.490
1000	Our Model	0.573	0.664	0.695
	BM25	0.362	0.433	0.498

3.2.2 Qualitative Analysis. To further verify the model’s effectiveness, we conducted qualitative analysis. For selected medical queries, the model was able to return high-quality, semantically relevant citation content. The generated passages displayed strong coherence and accuracy within the medical context, suggesting the potential practical value of our model beyond quantitative metrics.

3.2.3 Ablation Studies. We conducted ablation studies to investigate the impact of each component on the overall model performance. The results revealed that prefix optimization and relevance-based DocID construction play crucial roles in enhancing model performance. When these components were removed, the evaluation metrics (MRR@10, Success Rate @ 5, and Success Rate @ 10) significantly declined, further validating the efficacy of these design choices.

4 CONCLUSION

In conclusion, we have introduced our three-stage pipeline for the Biomedical Generative Retrieval (BioGen) task, which aims to enhance the reliability of large language models in the biomedical domain.

Our participation in this task has provided us with a deeper understanding of the potential of leveraging large language models to generate high-quality, contextually relevant biomedical responses. The fine-tuned T5 model for relevance-based DocID generation, combined with Residual Quantization for efficient document representation, has proven effective in improving retrieval quality, consistently outperforming traditional methods across various evaluation metrics.

Furthermore, our results demonstrate that our approach delivers contextually accurate and well-supported biomedical answers, indicating substantial advantages in leveraging generative retrieval for the biomedical domain. For our future work, we plan to explore the potential for improved performance by incorporating Incremental Learning techniques such as DSI++[8] and DynamicIR[9], as well as experimenting with other advanced retrieval optimization methods.

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